

Biodiversity Assessment for The Nature Conservancy

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Introduction:

This report evaluates an arid grassland and a mesic riparian wetland in order to recommend which habitat type The Nature Conservancy (TNC) should prioritize for conservation before high-density housing development. The purpose of this research is to use plant community data as an indicator of biodiversity, and then provide TNC with those data to supplement their consideration of each habitat as they plan for conservation efforts.

The riparian habitat is a federally listed wetland with a stream running through it and is generally the more scenic and recreationally enjoyed of the two sites. Due to its attractiveness to humans, this site is vulnerable to edge effects and exotic plant invasion. The grassland habitat is generally less scenic, but supports a rare wildflower that may be vulnerable to exploitation and harvesting, and it is home to ground-nesting birds that have few alternative spaces in the surrounding urban region. Although the highway intersecting the grassland generally means less aesthetic value, it also reduces foot traffic and limits some predator access into the site. Evidently, these two sites are both valuable but for different reasons; the riparian zone holds greater aesthetic value to the community as a beautiful public landscape, and the grassland holds significant ecological value due to its specialized habitat conditions and ability to support and help protect unique plant and wildlife species.

Methods:

Plant species were sampled within circular plots that were placed at random points along transects in both habitats. The species identification and percent cover were recorded for each detectable species (percent cover was used in place of the number of individuals as it is difficult to discern individuals in habitats containing many plants like those in our study). For the diversity calculations, percent cover values for each species were summed and then divided by the total number of plots sampled in that habitat.

To assess sampling effort and ultimately determine whether additional sampling would yield new species, a species accumulation curve was constructed by plotting the total number of species found on the Y axis and the number of plots sampled on the X axis. As shown in Figure 1, both habitats eventually reached a plateau, although the grassland required more plots to do so because it contained more total species. Three different measures of diversity were then

calculated: alpha diversity, referring to species richness or total number of species observed in each habitat, the Shannon-Wiener index (H), referring to how many species were present and how dominant or rare they were relative to each other, and evenness (E), referring to how evenly percent cover was distributed among the observed species.

Results:

The grassland habitat had greater species richness than the riparian habitat, with 13 species recorded across 30 plots, while the riparian habitat had 7 species recorded across 10 plots. The grassland also had a higher Shannon-Wiener diversity value ($H = 1.97$) compared to the riparian habitat ($H = 1.55$), but a slightly lower evenness ($E = 0.77$) compared to the riparian habitat (0.80) (Appendix A).

Table 1: Average Percent Cover of Species Observed in the Grassland and Riparian Habitats. Values are rounded from the original recorded values shown in Appendix A.

Species	Grassland Average % Cover	Riparian Average % Cover
<i>Convolvulus arvensis</i>	0.09	0.05
<i>Bromus tectorum</i>	0.46	—
<i>Liliaceae</i>	0.60	—
<i>Hordeum murinum</i>	0.08	—
<i>Taraxacum officianale</i>	0.14	0.10
<i>Bromus inermis</i>	0.53	—
<i>Chorispora tenella</i>	0.03	—
<i>Euphorbia seguieriana</i>	0.02	—
<i>Poa bulbosa</i>	0.05	—
<i>Asteraceae</i>	0.02	—
<i>Medicago sativa</i>	0.68	—
<i>Erodium cicutarium</i>	0.05	—
<i>Typha latifolia</i>	0.04	—
<i>Alpocurus pratensis</i>	—	0.63
<i>Rumex</i>	—	0.10
<i>Conium maculatum</i>	—	0.35
<i>Chemopodium</i>	—	0.05

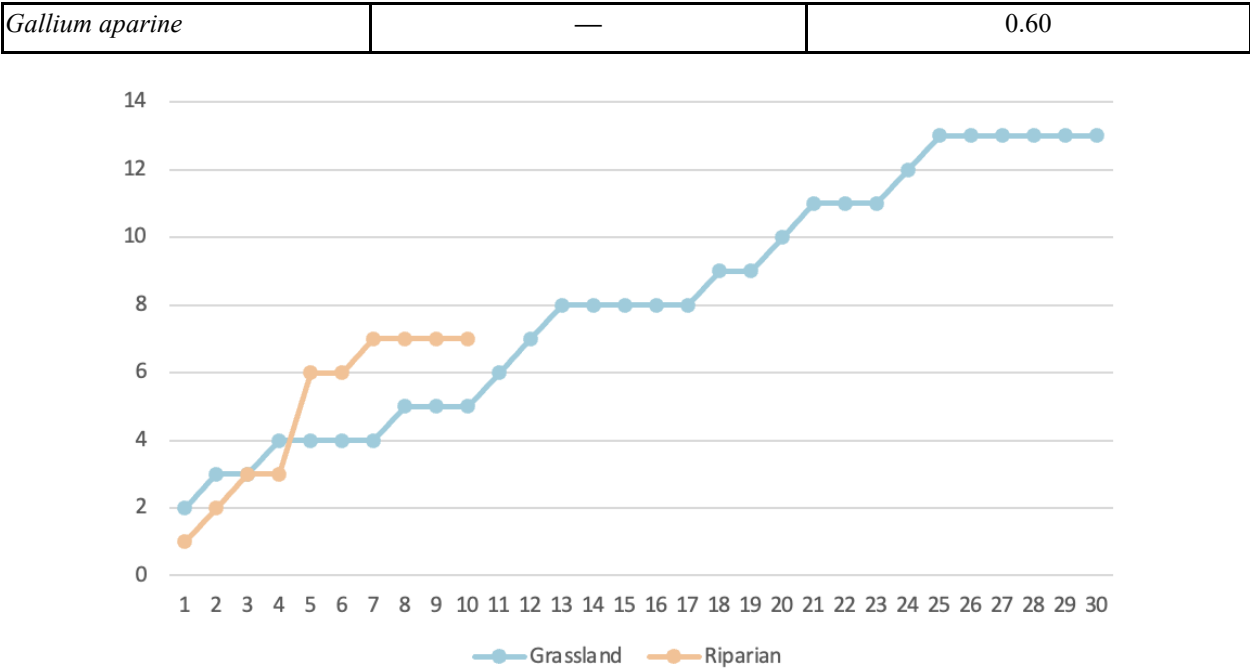


Figure 1: Species Accumulation Curves for the Grassland and Riparian Habitats.

The accumulation curves suggest that sampling was generally representative for both habitats (Figure 1). New species were observed in the grassland habitat through plot 25, with the final 5 plots yielding no new species. The grassland ultimately plateaued at 13 cumulative species. New species in the riparian habitat were observed through plot 7, with cumulative species remaining constant through plot 10.

Conclusion and Recommendation:

Based on both the qualitative information about the site and the quantitative biodiversity results, my overall recommendation is that The Nature Conservancy should prioritize the grassland habitat for conservation. Although the riparian habitat had slightly higher evenness, the grassland habitat had greater species richness and a higher Shannon-Wiener diversity value. These data overall point towards the grassland as being the more diverse plant community.

Because of its mission to “create a world where people and nature can thrive” (The Nature Conservancy 2019), I believe that TNC follows the land ethic whereby human

populations are considered part of the comprehensive ecosystem. Rather than being anthropocentric, this conservation ethic aims to “maintain the health of natural ecosystems and ecological processes” in the interest of all living beings, and while the grassland may not be the site that people tend to immediately recognize as being worth saving, the data and site characteristics indicate that it best protects diversity, rare species, and ecosystem functioning (Sher and González-Sargas 2026). Because of this, prioritization of the grassland site aligns best with TNC’s mission.

One limitation in this research is that percent cover values were estimates and may not be consistent among different observers. The sampling also only represents the plant community at one specific point in the season, so some species may have been left out if they were dormant or not yet growing/flowering at the time of sampling.

Overall, I recommend that TNC prioritize the grassland habitat for conservation. The riparian habitat is still valuable, but conserving the grassland would protect a site that supports greater biodiversity and ultimately better align with TNC’s mission.

References

Sher, A., and E. González-Sargas. 2026. An Introduction to Conservation Biology. Oxford University Press.

The Nature Conservancy. 2019. Who We Are | The Nature Conservancy. The Nature Conservancy. <https://www.nature.org/en-us/about-us/who-we-are/>.

Appendix

Appendix A: The following tables show the data and calculations for the Shannon-Wiener and evenness values. Average percent cover was calculated by dividing the percent cover for each species by the number of plots sampled per habitat, proportional abundance (pi) was calculated by dividing each species average percent cover by the total average percent cover for that habitat, Shannon-Wiener diversity was calculated by summing the values of proportional abundance * natural log of the proportional abundance, and evenness was calculated by dividing the Shannon-Wiener value by the natural log of the alpha diversity value.

Grassland

Species	Average % Cover	pi	ln(pi)	pi*ln(pi)
<i>Convolvulus arvensis</i>	0.086363636	0.03098403	-3.4742834	-0.1076473
<i>Bromus tectorum</i>	0.464285714	0.16656827	-1.79235	-0.2985486
<i>Liliaceae</i>	0.6	0.21525746	-1.5359205	-0.3306183
<i>Hordeum murinum</i>	0.08	0.02870099	-3.5508235	-0.1019122
<i>Taraxacum officianale</i>	0.1375	0.04932983	-3.0092262	-0.1484446
<i>Bromus inermis</i>	0.534210526	0.19165467	-1.6520601	-0.316625
<i>Chorispora tenella</i>	0.03	0.01076287	-4.5316528	-0.0487736
<i>Euphorbia seguieriana</i>	0.02	0.00717525	-4.9371179	-0.035425
<i>Poa bulbosa</i>	0.05	0.01793812	-4.0208271	-0.0721261
<i>Asteraceae</i>	0.02	0.00717525	-4.9371179	-0.035425
<i>Medicago sativa</i>	0.675	0.24216464	-1.4181375	-0.3434227
<i>Erodium cicutarium</i>	0.05	0.01793812	-4.0208271	-0.0721261
<i>Typha latifolia</i>	0.04	0.0143505	-4.2439707	-0.0609031

Total average percent cover = 2.787359877; Alpha diversity = 13; Shannon-Wiener index = 1.971997813; Evenness = 0.768825243.

Riparian

Species	Average % Cover	pi	ln(pi)	pi*ln(pi)
<i>Alpoceurus pratensis</i>	0.62777778	0.33431953	-1.0956581	-0.3662999
<i>Rumex</i>	0.1	0.05325444	-2.9326741	-0.1561779
<i>Conium maculatum</i>	0.35	0.18639053	-1.6799112	-0.3131195
<i>Chemopodium</i>	0.05	0.02662722	-3.6258213	-0.0965455
<i>Gallium aparine</i>	0.6	0.31952663	-1.1409147	-0.3645526
<i>Taraxacum officianale</i>	0.1	0.05325444	-2.9326741	-0.1561779
<i>Convolvulus arvensis</i>	0.05	0.02662722	-3.6258213	-0.0965455

Total average percent cover = 1.87777778; Alpha diversity = 7; Shannon-Wiener index = 1.54941894;
Evenness = 0.79624383.